Parity Blindness of Finite-Window Weil Detectors at a Real Zero: the Even Sector Cannot See a Siegel Configuration and the Odd Sector Is Maximally Sensitive to It

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Abstract

In the finite-window Weil form, a real zero pair $\{\frac{1}{2}+\delta, \frac{1}{2}-\delta\}$ contributes $2F(i\delta)F(-i\delta)$, and because the autocorrelation factorises this is $+2A_\delta(f)^2$ for even test functions and $-2S_\delta(f)^2$ for odd ones, with $A_\delta = \int f\cosh(\delta u)\,du$ and $S_\delta = \int f\sinh(\delta u)\,du$. The perturbation is therefore rank one, positive semidefinite on the even sector and negative semidefinite on the odd sector, for every window length and in every basis. Two consequences follow. Even-sector finite-window positivity is compatible with a real zero at every $\delta \in (0, \frac{1}{2})$, so no even-sector positivity computation — at any L, in any basis, to any precision — carries information about the Siegel configuration; the window is not merely blind there, it is anti-detecting, and the statement is exact in δ rather than a vanishing leading coefficient. Conversely the odd-sector sensitivity at $\gamma = 0$ is the γ-uniform maximum $2 \cdot (2L^3/3)$ of the identity $K_{\text{even}}(L,\gamma) + K_{\text{odd}}(L,\gamma) = 2L^3/3$, so $\gamma = 0$ is the unique ordinate at which the two sectors' roles separate completely. The sign structure is verified in 150 cells to 8.49×10^{-61}, reproduces the repository's independently derived odd-sector pole term entrywise with no normalisation adjustment, and holds exactly over $\mathbb{Q}$ in the function-field control. This resolves non-claim 9 of the companion note [2].

0. Non-claims

Stated first.

1. **Nothing about RH or GRH**, and **nothing about whether Siegel zeros exist.** The theorem below describes what a window form does when a real zero is placed in the zero multiset by hand. It is a statement about a detector, not about the zeros of any L-function.
2. **The quasi-Riemann-hypothesis and Siegel-exclusion preprints of October 2026 are not cited as established and are not used anywhere.** No result here depends on them, and none is asserted about them.
3. **Every planted detection is [T2-N].** A planted real zero is a perturbation of a computed matrix, not an arithmetic fact, and **non-detections certify nothing**.
4. **"The odd sector detects by $L = 1.4$" is [T2-N] twice over.** It is a numerical statement about a planted configuration, and the odd-sector *baseline* it is measured against has no

certified lower bound: no Zhu-style certificate exists for any Dirichlet window. Converting the measurement into an exclusion of a real zero would require exactly such a bound, and this note supplies none.

5. **The parent extension's Dirichlet-character results beyond its §5 are not used here.** Saturation ratios, prime-side coincidences, conductor-versus-height observations and the Phase-4 mechanism hypothesis play no part in anything below.
6. **Corollary 2 bounds what a method can detect, not what exists.** Section 5 is a scope statement about even-sector techniques. It is not a criticism of any result, and no cited work claims otherwise.
7. **The function-field section is a control, not a transfer.** In the function field the analogue of RH is Weil's theorem, so there is no configuration there to exclude.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T2-C	certified arithmetic	an interval/ball enclosure with a reproducible certificate
T2-N	high-precision numerical	stable under precision increase, not an enclosure
T3	interpretation	argument from T1/T2 evidence, labelled as such

Every number printed in this note is re-derived from the extension's data files by `note/verify_note_numbers.py`, which also fails if the note prints more significant digits than its source carries.

1 Setup

Let g be real and supported in $[-L, L]$. Following the companion note [2, §2], put

$$G(z) = \int g(u)e^{zu}\,du, \qquad h(v) = \int g(u+v)g(u)\,du, \qquad H(z) = G(z)G(-z),$$

so h is supported in $[-2L, 2L]$ and $H(it) = |G(it)|^2$ for real g. The spectral side of the explicit formula runs over the zeros of the completed function, and all reported values are normalised as $Q(g)/\|g\|_2^2$. For ζ the window form is this repository's assembled Q_ζ; for a real primitive χ modulo q it is

$$M_\chi = \mathrm{Arch}_a - \mathrm{LogPi} + \mathrm{LogQ} - \mathrm{Prime}_\chi,$$

with no pole term, χ being non-principal. Both parity sectors are used throughout: the even bases $\cos((2k+1)\pi x/2L)$ and $\cos(k\pi x/L)$, and the odd basis $\sin(k\pi x/L)$, all normalised.

For a non-real off-line zero the complete functional-equation and conjugation orbit is $\mathcal{O} = \{\beta \pm i\gamma,\ 1-\beta \pm i\gamma\}$, and its contribution is [2, eq. (4)]

$$Q_{\mathcal{O}}(g) = 4\,\mathrm{Re}\Big(A\overline{B}\Big) = 4\left(|P|^2 - |S|^2\right), \qquad A = G(\delta+i\gamma), \quad B = G(-\delta+i\gamma), \tag{1}$$

with $P = \int g\cosh(\delta u)e^{i\gamma u}du$ and $S = \int g\sinh(\delta u)e^{i\gamma u}du$, $\delta = \beta - \frac{1}{2}$: a signed rank-two form, positive on the cosh channel and negative on the sinh channel. The present note is the $\gamma = 0$ corner of (1), where the orbit degenerates to a *pair* and the two channels separate completely by parity.

2 The theorem

Theorem 1 (sign structure at a real zero)**.** *[T1] Let f be real and supported in $[-L, L]$, and let a real zero pair $\{\frac{1}{2}+\delta,\ \frac{1}{2}-\delta\}$, $\delta \in (0, \frac{1}{2}]$, be present in the zero multiset. Its contribution to the finite-window Weil form is*

$$2\left(\int f(u)e^{\delta u}du\right)\left(\int f(u)e^{-\delta u}du\right) = \begin{cases} +2\,A_\delta(f)^2, & f \text{ even}, \quad A_\delta(f) = \int f(u)\cosh(\delta u)\,du, \\ -2\,S_\delta(f)^2, & f \text{ odd}, \quad S_\delta(f) = \int f(u)\sinh(\delta u)\,du, \end{cases}$$

for every $L > 0$ and in every basis. In particular the perturbation is rank one, positive semidefinite on the even sector and negative semidefinite on the odd sector.

Proof. Four lines; written out with every step in Appendix A. The zero side is $\sum_\rho h(z_\rho)$ with $h(z) = F(z)F(-z)$, $\rho = \frac{1}{2} + iz$, and the autocorrelation *factorises* — that is the whole content. The functional equation forces the partner, so the pair sits at $z = \mp i\delta$ and contributes $h(i\delta) + h(-i\delta) = 2h(i\delta) = 2F(i\delta)F(-i\delta)$. With $F(i\delta) = \int f(x)e^{-\delta x}dx$ and $e^{-\delta x} = \cosh(\delta x) - \sinh(\delta x)$, parity gives $F(\pm i\delta) = (A_\delta, A_\delta)$ for even f and $(-S_\delta, +S_\delta)$ for odd f. Even f gives $A \cdot A$, odd f gives $S \cdot (-S)$. The sign is not a convention: it is the parity of F at a purely imaginary argument. □

Corollary 2 (the even sector is anti-detecting)**.** *[T1] Suppose every non-trivial zero of $L(s, \chi)$ lies on the critical line except a single real pair $\frac{1}{2} \pm \delta$. Then for every $L > 0$ and every real even f supported in $[-L, L]$,*

$$Q_\chi(f) = \sum_{\text{on-line } \rho} |F(\gamma_\rho)|^2 + 2A_\delta(f)^2 \ \geq\ 2A_\delta(f)^2 \ \geq\ 0.$$

Hence even-sector finite-window Weil positivity is compatible with a real zero at every $\delta \in (0, \frac{1}{2})$, and no even-sector positivity computation — at any window length, in any basis, to any precision — carries information about the Siegel configuration. Equivalently, as a perturbation of the matrix: the response is $+2aa^{\mathsf{T}}$ with $a_k = \int \varphi_k \cosh(\delta u)\,du/\|\varphi_k\|$, so $\lambda_{\min}(Q + 2aa^{\mathsf{T}}) \geq \lambda_{\min}(Q)$ for every basis and every N.

Remark 3. [T1] This is stronger than "the δ^2 coefficient vanishes". That weaker statement is true — $K(L, 0) = \int_{-L}^{L} u^2 \sin^2(0 \cdot u)du = 0$, i.e. $F'(0) = 0$ for even f — but it only kills the *leading* coefficient. Theorem 1 is exact in δ: the response is a perfect square of the wrong sign to all orders. The even sector is not blind at $\gamma = 0$; a planted real zero makes the form *more* positive.

Corollary 4 (the odd sector is maximally sensitive)**.** *[T1] For real odd f the same identity reads $Q_\chi(f) = \sum_{\text{on-line}} |\widetilde{F}(\gamma_\rho)|^2 - 2S_\delta(f)^2$, which can be negative. Moreover the derivative-evaluation extremal in the odd sector at $\gamma = 0$ is*

$$2\sup\left\{\left(\int_{-L}^{L} u\,g(u)\,du\right)^2 : g \text{ odd},\ \|g\|_2 = 1\right\} = 2\,K_{\text{odd}}(L, 0) = 2 \cdot \frac{2L^3}{3},$$

attained at $g \propto u$, which is the γ-uniform maximum *of the identity $K_{\text{even}}(L, \gamma) + K_{\text{odd}}(L, \gamma) = 2L^3/3$ together with $K_{\text{even}}(L, 0) = 0$.*

So $\gamma = 0$ is a blind spot of one *sector*, not of the window, and the complementary sector is there as sensitive as it ever gets at any ordinate.

Remark 5 (general f: lowering the form needs an odd component). [T1] For real $f = f_{\text{e}} + f_{\text{o}}$ the cross integrals vanish by parity, so the computation of Appendix A gives $F(i\delta) = A_\delta(f_{\text{e}}) - S_\delta(f_{\text{o}})$ and $F(-i\delta) = A_\delta(f_{\text{e}}) + S_\delta(f_{\text{o}})$, and the real pair contributes $2(A_\delta(f_{\text{e}})^2 - S_\delta(f_{\text{o}})^2)$. A real pair can therefore lower the form only through a test function with a nonzero odd component, and only when $|S_\delta(f_{\text{o}})| > |A_\delta(f_{\text{e}})|$. The statements below are in this sense about the odd sector itself and not about a choice of basis.

$\gamma = 0$ **is the only ordinate where the sectors separate.** The identity $K_{\text{even}} + K_{\text{odd}} = 2L^3/3$ holds for every γ, but the split is extreme only at $\gamma = 0$. At large γ the two are nearly equal: the companion note measures $K_{\text{even}}/K_{\text{odd}} = 1.00634$ at $L = 2$ and $\gamma = \gamma_1$, the first off-line ordinate of the Davenport–Heilbronn function, $\gamma_1 \approx 85.7$ [2, §6]. The complete separation of roles is a $\gamma = 0$ phenomenon.

Verified at $\gamma = 0$, with K_{odd} obtained by quadrature rather than by subtraction:

L	$K_{\text{even}}(L, 0)$	$K_{\text{odd}}(L, 0)$	$2L^3/3$	|err|
0.5	0	0.0833333333333	0.0833333333333	0
0.8	0	0.341333333333	0.341333333333	0
1.0	0	0.666666666667	0.666666666667	0
1.3	0	1.46466666667	1.46466666667	0
2.0	0	5.33333333333	5.33333333333	0

3 Verification

Three independent checks, from `src/verify_siegel_parity.py`. All three passed, and were re-run for this note and reproduced the committed transcript byte for byte.

(a) The planted object was the real *pair*. Decided from the committed Phase-5 data rather than from a reading of the code, by recomputing stored eigenvalues both ways. The degenerate quartet is exactly twice the pair — $\max|\text{quartet}(\gamma = 0) - 2 \cdot \text{pair}| = 4.98 \times 10^{-60}$ — so the two hypotheses differ by a factor of 2 and the data distinguishes them: all twelve probed rows are reproduced by the PAIR, unanimously. **No factor-of-2 correction is needed anywhere in this note.** [T2-N]

(b) Rank one, correct sign, eigenvalue $= +2A^2$ **/** $= -2S^2$**.** 5 characters $\times$ $L \in \{0.8, 1.0, 1.4\}$ $\times$ 2 sectors $\times$ $\delta \in \{0.1, 0.25, 0.375, 0.45, 0.49\}$, $N = 10$: 150 cells, 150 with the predicted sign. A and S are computed by quadrature on the explicit normalised eigenfunction, so this is an independent check of the closed forms, not a restatement of them. [T2-N]

quantity	worst over the 150 cells
relative error, eigenvalue vs $+2A^2$ (even) / $-2S^2$ (odd)	8.49×10^{-61}
rank-one residual $\max_{ij} \lvert B_{ij} - \lambda v_i v_j \rvert$	2.49×10^{-60}
largest non-dominant eigenvalue	1.45×10^{-60}
the complementary integral (S on even f, A on odd f)	exactly 0
even-sector eigenvalues, 75 of 75	$+3.14163 \ \ldots \ +6.38933$
odd-sector eigenvalues, 75 of 75	$-0.905634 \ \ldots \ -0.00643962$
$K_{\text{even}} + K_{\text{odd}} = 2L^3/3$, K_{odd} by quadrature	1.56×10^{-61}

L	δ	$+2A_\delta^2$ (even)	$-2S_\delta^2$ (odd)	rank-one resid.	rel. err.
0.8	0.1	3.14162843133	−0.00643961708992	6.22×10^{-61}	8.49×10^{-61}
0.8	0.49	3.29375936732	−0.159042352281	3.69×10^{-61}	1.89×10^{-61}
1.0	0.1	3.93155834205	−0.0125860661447	6.42×10^{-61}	1.58×10^{-61}
1.0	0.49	4.23383664019	−0.315813694881	6.22×10^{-61}	2.46×10^{-61}
1.4	0.1	5.52110947823	−0.0345998225163	6.22×10^{-61}	5.64×10^{-61}
1.4	0.49	6.38932656805	−0.905634273488	6.22×10^{-61}	2.58×10^{-61}

(The extreme δ only; all 150 cells are in `data/verify_siegel_parity.csv`.)

The matrices do not depend on χ at all. Grouping the 150 cells by $(L, \text{sector}, \delta)$ leaves exactly *one* distinct eigenvalue in each of the 30 groups: the response to a real zero is a property of the basis and δ only. That is why the conclusion is uniform over the family rather than five separate measurements, and it is why the table above has no χ column.

(c) Pole consistency — the strongest check in this note. The $\delta = \frac{1}{2}$ real-pair perturbation was compared entrywise with this repository's ζ-side pole term $2\,\mathrm{pole_sign(sector)} P_j P_k / \|\cdot\|$, reporting *both* differences so that the data fixes the sign. [T2-N]

sector	L	pole sign	$\max\lvert B - \mathrm{Pole}\rvert$	$\max\lvert B + \mathrm{Pole}\rvert$	vanishes
EVEN	0.5	+1	1.07×10^{-61}	4.084	B − Pole
EVEN	0.8	+1	3.11×10^{-61}	6.749	B − Pole
EVEN	1.3	+1	3.6×10^{-61}	11.95	B − Pole
EVEN_D	0.5	+1	1.94×10^{-62}	3.281	B − Pole
EVEN_D	0.8	+1	3.11×10^{-61}	5.347	B − Pole
EVEN_D	1.3	+1	6.22×10^{-61}	9.128	B − Pole
ODD	0.5	−1	1.22×10^{-63}	0.05108	B − Pole
ODD	0.8	−1	9.72×10^{-63}	0.2119	B − Pole
ODD	1.3	−1	1.94×10^{-62}	0.9408	B − Pole

$B = \mathrm{Pole}$ exactly in all 9 cells, with $\max\lvert B - \mathrm{Pole}\rvert \le 6.22 \times 10^{-61}$, while the alternative $\max\lvert B + \mathrm{Pole}\rvert$ is 0.05108 to 11.95 — wrong by $O(1)$ to $O(10)$, so the test is decisive. **There is no normalisation mismatch: no factor of 2 and no factor of $\frac{1}{2}$.** This is the strongest check because the pole term was derived independently, for a different purpose, years-of-work earlier in this repository's own ζ assembly, and Theorem 1 reproduces it as the $\delta = \frac{1}{2}$ instance of $-2S_\delta^2$ in the odd sector and of $+2A_\delta^2$ in the even ones.

Remark 6 (a sign that must not be misread). The odd sector's pole_sign is -1, so the odd-sector pole term carries the sign *opposite to the even-sector one*, and the real-pair perturbation reproduces that flip exactly. It is **not** opposite in sign to the pole term itself. The distinction matters because Corollary 2 rests on this sign.

The χ-analogue is identically zero, and that is the point. M_χ reconstructs as $\mathrm{Arch} - \mathrm{LogPi} + \mathrm{LogQ} - \mathrm{Prime}$ with residual exactly 0: for non-principal χ there is no pole term at all. Meanwhile the real-pair perturbation at $\delta = \frac{1}{2}$ is *not* zero for χ (largest entry 2.673477). On the ζ side a pole-like object at $z = \mp i/2$ is already present in the form; on the χ side there is nothing of the kind to cancel against a real zero.

4 The function-field analogue

[T1, exact over $\mathbb{Q}$] The same configuration exists in this repository's function-field control, where RH is Weil's theorem. There a real reciprocal pair $\{\rho, \rho^{-1}\}$ gives the rank-two Toeplitz matrix of Theorem 6 of the δ^2–L^3 paper [3],

$$T_R = uw^{\mathsf{T}} + wu^{\mathsf{T}}, \qquad u_i = \rho^i, \quad w_i = \rho^{-i}, \quad 0 \le i \le R.$$

With J the flip $x_i \mapsto x_{R-i}$ one has $Ju = \rho^R w$ and $Jw = \rho^{-R} u$, so the parity directions are the **ρ^R-weighted** vectors $v_\pm = u \pm \rho^R w$, and

$$T_R = \frac{1}{2\rho^R}\left[v_+ v_+^{\mathsf{T}} - v_- v_-^{\mathsf{T}}\right], \qquad T_R v_\pm = \lambda_\pm v_\pm, \qquad \lambda_\pm = (R+1) \pm \rho^{-R}\|u\|^2.$$

Verified exactly in `fractions.Fraction`, no floating point, for $\rho \in \{2, \frac{3}{2}, \frac{5}{4}, \frac{7}{3}, 1\} \times R \in \{1, 2, 4, 8, 16\}$ — 25 cells, 0 failures: the splitting identity, both eigenvector relations, v_+ palindromic and v_- anti-palindromic under $i \mapsto R - i$, $\lambda_+ > 0$, and $\lambda_- \le 0$ with equality iff $\rho = 1$. The negative eigenvalue, exactly:

ρ	$R=1$	$R=2$	$R=4$
2	$-1/2$	$-9/4$	$-261/16$
3/2	$-1/6$	$-25/36$	$-5125/1296$
5/4	$-1/20$	$-81/400$	$-168561/160000$
7/3	$-16/21$	$-1600/441$	$-6088000/194481$
1	0	0	0

($R=8$ and $R=16$ are verified in the same run; their exact rationals run to 22 digits in the denominator and are not printed.)

Remark 7 (a correction, carried over). The brief that commissioned the parent extension, and the brief that commissioned this note, both state that the eigenvalues $(R+1)\pm\|u\|\|w\|$ are attained on $u+w$ and $u-w$, with $u+w$ palindromic. That is **false for every** $\rho\neq 1$: it holds in exactly 5 of the 25 cells, which are precisely the degenerate $\rho=1$ ones, and $T_R(u+w)$ is parallel to $u+w$ only when $\rho^{2R}=1$. The missing factor is ρ^R. The *polarisation identity* $T_R=\frac{1}{2}[(u+w)(u+w)^{\mathsf{T}}-(u-w)(u-w)^{\mathsf{T}}]$ is nevertheless exact in all 25 cells. Neither correction touches any result of [3], whose proof derives the eigenvalues basis-independently and never names $u\pm w$ as eigenvectors.

What the control shows. The J-even direction carries the positive rank-one piece and the J-odd direction the negative one: the same signed split, with the same attachment of sign to parity, as $+2A_\delta^2$ / $-2S_\delta^2$. So the sign structure is intrinsic to "a real pair in a finite window" and is not an artefact of the archimedean bookkeeping of $L(s,\chi)$. This is a check at a level the leading-constant dictionary of [3] did not reach: that dictionary matched a δ^2 coefficient between the two sides, whereas what matches here is the *sign attached to a parity class*, an exact and discrete datum. It does **not** transfer the consequence of Corollary 2: in the function field RH is a theorem, so there is no configuration to exclude.

5 Consequences for even-sector techniques

[T3] This section is a scope statement about what certain methods can detect. It is not a criticism of any result, and none of the work cited claims otherwise.

Theorems 7, 8 and 11 of the δ^2–L^3 paper [3] are labelled even-sector in their own statements, and that paper says so globally. By Corollary 2, the even sector is precisely the one that cannot respond to a real zero: a positivity statement there is compatible with a real zero at any $\delta\in(0,\frac{1}{2})$. So each of those three is silent on the Siegel configuration *by construction* — not for want of window length, basis size or precision, but because the perturbation it would have to see is positive semidefinite in that sector. The same applies to the four-mode certified witness of Theorem 11, whose witness consists of even quarter-wave modes.

The Connes–van Suijlekom pipeline [1] enters the same way at one specific point: its hypothesis concerns a simple lowest eigenvalue with an *even* eigenfunction, so the quantity it is about is an even-sector ground state. Corollary 2 says that quantity cannot move in response to a planted real zero.

Zhu's certificate is a different case, and this note does not settle it. It would be convenient to add Zhu's $\lambda^*(0.8)\geq 8.9\times 10^{-18}$ [5] to the list. That does **not** follow, and nothing here establishes it either way, because the *domain* of Zhu's infimum is deliberately left open by [3], which says only that a certificate proved on the full admissible space may be quoted unchanged. The two cases differ in kind:

- if $\lambda^*(L)$ is an infimum over the **even sector only**, then by Corollary 2 it carries no information about a real zero, exactly as for Theorems 7, 8 and 11;

- if it is an infimum over the **full admissible space**, then it *does* constrain one, since positivity on the full space includes the odd sector where the response is $-2S_\delta(f)^2$ — and that would be the opposite conclusion, and the more interesting one.

Which case holds is a property of a paper that was not obtained or read for this work. **No claim is made in either direction.** The brief that commissioned this note listed Zhu's certificate among the even-sector cases; that listing is not supported, and is corrected here rather than repeated.

Where a window-based statement about real zeros would have to come from. From a test function with a nonzero odd component, and from nothing else: by Remark 5 a real pair contributes $2(A_\delta(f_\mathrm{e})^2 - S_\delta(f_\mathrm{o})^2)$, so lowering the form requires the odd part to dominate. The odd sector at small L is where that is easiest, the response there being negative and largest relative to the ordinate (Corollary 4). But the route is subject to the obstruction the companion note [2] measured on the Davenport–Heilbronn function: there, the negativity produced by an off-line orbit was partly cancelled by the remaining spectrum, and turning a detector response into an exclusion needs an unconditional lower bound on the remaining-zero form. No such bound is known to the author for any Dirichlet window, and none is produced here (non-claim 4) — a statement about what is available, not a proof that none exists.

6 Illustration

[T2-N; every row] Smallest scanned L at which a planted real zero at each δ produces a negative odd-sector Rayleigh quotient, at fixed basis dimension $N = 12$, over $L \in \{0.5, 0.8, 1.1, 1.4, 1.7, 2.0, 2.3, 2.6, 3.0\}$ (270 rows, 228 detections):

	$\delta = 0.1$	$\delta = 0.25$	$\delta = 0.375$	$\delta = 0.45$	$\delta = 0.49$
ζ	0.5	0.5	0.5	0.5	0.5
χ_{-3}	0.8	0.8	0.8	0.8	0.8
χ_{-4}	1.1	1.1	1.1	0.8	0.8
χ_5	1.1	1.1	0.8	0.8	0.8
χ_{-7}	1.4	1.1	1.1	1.1	1.1
χ_8	1.4	1.1	1.1	1.1	1.1

So: **odd sector, by $L = 1.4$ with a 12-dimensional basis, for every character and every $\delta \geq 0.1$**; as early as $L = 0.5$ for ζ and $L = 0.8$ for χ_{-3}. And **even sector: never**, at any L, any N, any δ — in the 180 even-sector rows of the Phase-5 scan, 180 have response ≥ 0 and 0 are detections, while all 180 odd-sector rows have response < 0 and 135 are detections. That is Theorem 1 as a prediction, confirmed, not as evidence for itself.

Remark 8 (two caveats that change how the table should be read). First, the ζ row is nearly vacuous: the odd-sector baseline there is $\lambda^*_{\mathrm{odd}}(\zeta, 0.8) = 4.98077 \times 10^{-14}$, so ζ's odd form is already within 10^{-14} of singular and *any* negative rank-one perturbation detects. The informative rows are χ_{-4} and χ_5 at $L = 0.8$, which clear baselines of 0.0988 and 0.0679 respectively (both at $N = 12$). Second, the ordering of the onsets is the ordering of the odd-sector baselines, and across these five characters the conductor q and the first ordinate γ_1 are not independently varied, so the ordering is not evidence of a conductor effect.

7 What this rules out, and what it does not

Ruled out. Reading any even-sector finite-window positivity result — at any L, in any basis, to any precision — as evidence against a real zero near 1. By Theorem 1 the perturbation such

a result would have to detect is positive semidefinite in that sector, so the computation is not insensitive to the configuration but *anti*-sensitive to it: planting the zero improves the very quantity being certified.

Not ruled out, and not claimed. Whether any real zero near 1 exists, for ζ or for any $L(s,\chi)$: nothing here bears on it. An odd-sector certificate: none is given, and none is known for any Dirichlet window, so the odd-sector results are detector-sensitivity measurements and not exclusions. Complex characters: the whole analysis assumes χ real, since it is the reality of χ that makes M_χ real symmetric and the parity split available. And the masking obstruction of the companion note is untouched — this note says where a real zero would be visible, not that it could be isolated from the rest of the spectrum once it was.

A The proof, with the factorisation written out

The content of Theorem 1 is that the autocorrelation factorises. Writing it out once, with every step:

$$\text{zero side} = \sum_\rho h(z_\rho), \qquad h(z) = F(z)F(-z), \qquad \rho = \tfrac{1}{2} + iz,$$

$$\text{real pair } \{\tfrac{1}{2} \pm \delta\} \;\longrightarrow\; z = \mp i\delta \qquad \text{(the functional equation forces the pa}$$

$$\text{contribution} = h(i\delta) + h(-i\delta) = 2h(i\delta) \qquad (h \text{ even})$$

$$= 2\,F(i\delta)\,F(-i\delta) \qquad \text{(the factorisation — the whole conten}$$

$$F(i\delta) = \int f(x)e^{-\delta x}dx = \int f\cosh(\delta x)dx - \int f\sinh(\delta x)dx$$

$$f \text{ even}: \quad A_\delta - 0 = A_\delta, \qquad F(-i\delta) = A_\delta + 0 = +A_\delta$$

$$f \text{ odd}\ : \quad 0 - S_\delta = -S_\delta, \qquad F(-i\delta) = 0 + S_\delta = +S_\delta$$

so the product is $+A_\delta^2$ for even f and $-S_\delta^2$ for odd f, and the contribution is twice it. In matrix form, with $a_k = \int \varphi_k \cosh(\delta u)du/\|\varphi_k\|$ and $s_k = \int \psi_k \sinh(\delta u)du/\|\psi_k\|$ on the even and odd bases, the perturbations are $+2aa^{\mathsf{T}}$ and $-2ss^{\mathsf{T}}$: rank one, of the stated signs.

Two points worth isolating. The sign is *not* a convention: it is the parity of F at a purely imaginary argument, and $F(\pm i\delta)$ are both real for real f of definite parity, with the *same* sign in the even case and *opposite* signs in the odd case. And the statement is exact in δ: nothing is expanded, so there is no $\delta \to 0$ regime in which it fails.

B Retraction log

Reproduced from the parent extension's §0.2, verbatim in substance, plus this session's.

R1. `src/regression_zeta.py` asserted the ζ-side values against the parent's committed CSVs at relative tolerance 10^{-20} and reported 14 failures. **Retracted: the tolerance was wrong, the values are right.** The parent prints eigenvalues to twelve significant digits, so a full-precision recomputation can agree with the stored *decimal string* only to about 10^{-12}; every "failure" sat at $4 \times 10^{-15} \ldots 1.7 \times 10^{-12}$. Tolerance corrected to 5×10^{-12}. No computed value changed. The binding precision on any regression against this repository's CSVs is the CSVs' own.

R2. `src/phase2_gateA_primeflip.py` built $L =$ `mpf('0.8')` *before* setting the working precision, so L carried only 15 digits into a 40-digit run. **Fixed.** The affected quantities were control ratios of order 10^4, so no reported conclusion moved, but the ordering was wrong and is recorded rather than quietly repaired.

R3. `src/planting.py::quartet_block` initially omitted the factor $h_{\text{sign}} = -1$ for the odd basis, where the repository's F returns F/i. **Fixed before use**, caught by the planting self-test (quartet$(\gamma, 0) = 2 \cdot$ pair(γ) failed by 4.04). No result was computed with the bad version.

R4. A commit message listed a fourth retraction — the ζ Connes window — but the row was never added to the table. **Fixed.** The substance: that file's fourth window is $L = \log(13)/2$ stored rounded to 10 digits, and $\lambda_{\min} \approx 10^{-29}$ there is hypersensitive because $2L$ sits exactly at the $n = 13$ prime-power cutoff, so reconstructing L from the stored string moved the answer by 1.8×10^{-8} relative. *The documentation gap is itself the retraction: a commit message is not the record.*

R5. `src/verify_siegel_parity.py` first asserted a stored column at relative tolerance 10^{-25}, which would have reported 12 spurious failures at observed deviations of $4 \times 10^{-11} \ldots 1.3 \times 10^{-10}$. **Fixed before the run that produced the committed transcript**: the stored column carries ten significant digits, so the binding tolerance is the file's. *This is R1 recurring — the lesson did not transfer the first time.*

R6 (this session). Two claims in the brief that commissioned this note were repeated from an earlier brief after the parent extension had already refuted them, and are **not** asserted here: that the function-field eigenvalues are attained on $u \pm w$ with $u + w$ palindromic (false for every $\rho \neq 1$; Remark 7), and that Zhu's certificate is one of the even-sector cases that carry no information about the Siegel configuration (not established either way; §5). Both are corrected in place rather than carried forward.

C Reproduction

Code and data: `https://github.com/innerlightr-wq/weil-window-control`, directory `extensions/dirichlet`
Python 3.14.7 (CPython), `mpmath` 1.3.0; no `numpy`, no `float64` on any path to a reported number;
§4 uses `fractions.Fraction` only.

Result	Script	Data
§3 checks (a), (b), (c)	`src/verify_siegel_parity.py`	`data/verify_siegel_parity.csv`, `results/verify_siegel_pari`
§2 the K identity	`src/phase4b_sector_extremal.py`	`data/phase4b_sector_extremal.csv`
§4 function field	`src/ff_parity_split.py`	`results/ff_parity_split.json`
§6 onset, odd sector	`src/phase5_onset.py`	`data/phase5_onset_odd.csv`
§6 even/odd scan	`src/phase5_siegel.py`	`data/phase5_siegel.csv`

`make verify`, in the note directory, runs `note/verify_note_numbers.py`: it rebuilds every quantity printed above from the data files, scans this document for numeric literals, fails if any literal is not derived or is printed to more significant digits than its source carries, regenerates the five machine-generated tables and compares them, and re-checks that the three prerequisite verifications are recorded as passed. `make` builds the PDF. The full extension suite is `./run_suite.sh`.

Licence

As the parent repository: code MIT, this note and the data CC BY 4.0. CC BY 4.0 requires attribution; if you use the note or the data, please cite it by its DOI and the parent paper [3] separately.

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